



MASTERSCAN ENGINEERING

INSPECTION WITH INTEGRITY

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GST/Co. Reg. No.: 200506867D 29 Benoi Road, Pioneer Lot, Singapore 629922
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POSITIVE MATERIAL IDENTIFICATION

CLIENT	: INVX Asia Pacific Pte.Ltd.	REPORT NO	: SFO-2886	PAGE NO : 1 OF 3
SITE LOCATION	: 80 TUAS WEST DRIVE, SINGAPORE 638417	MSE/REF/NO	: D2-26-2389	
PO NUMBER	: 3759-00	TESTED DATE	: 30.03.2026	
PROJECT NAME	: SHELL MALIKAI	TOTAL POINTS	: 03	
SALES ORDER	: S54000355	PMI INSTRUMENT MANUFACTURER	: THERMO SCIENTIFIC	
INSPECTION AND TEST PLAN NUMBER	: QCP10116.07.01, 10116.02.02, 10116.02.05, 10116.02.06, 10116.02.09 and 10116.02.12	MODEL	: XL2 -100G	
PART NUMBER	: 2-702549-02	SERIAL NO	: 96976	
SERIAL NUMBER (SFO)	: 2886	CALIBRATION DATE	: 07/01/2026	
DESCRIPTION OF EVENT	: WITNESS MATERIAL PMI	CALIBRATION DUE DATE	: 07/01/2027	
TEST PROCEDURE	: QP-10-43 Rev:D / ASTM E-1476			
ITEM TESTED	: REFER ATTACHED SHEET	SOURCE TYPE	: X-RAY TUBE	

Sr.No	Description	Part Number	Serial Number (SFO)	Reading No	Item	Design Material	PMI Result	Cr	Ni	Mo	Nb	Ti	Acceptance (Yes/No)
	ACCEPTANCE CRITERIA : DQM-0303 Rev'E												
	PRE VERIFICATION FOR KNOWN MATERIAL												
				2576	P.NO : BS625C	Alloy 625	Alloy 625	20.0-23.0	58.0Min	8.00-10.00	3.15-4.15	0.40Max	
								23.26	60.25	8.75	3.58	0.32	Yes

NOTE :

SUBJECT (AS RECEIVED) : One (01) unit of steel pipe spool test sample, marked as stated below, for positive material identification (PMI) Test. Figure below shows the tested locations.

TEST RESULT : The alloy and Chemical Composition of the materials are shown in the attached result sheets.

Tested and Reported By : M.KAJENDRAN NDT LEVEL II	Reviewed By : SURENDRA KUMAR P.V ASNT NDT Level III	Client Reviewed By : INNOVEX	Owner Received By :
Signature	Signature	Signature	Signature
Date : 30.03.2026	Date : 31.03.2026	Date :	Date :

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Sr.No	Description	Part Number	Serial Number (SFO)	Reading No	Item	Design Material	PMI Result	Cr	Ni	Mo	Nb	Ti	Acceptance (Yes/No)
ACCEPTANCE CRITERIA : DQM-0303 Rev:E													
1	GVS, GATE MANUAL, 2.06", RATED PRESSURE 5000 PSI, TYPE DH, API 6A, U, HH-NL, PSL-3, CRA2, TCHF, SPECIAL, SHELL MALIKAI QUALITY PLAN DQAPQCP 10116.02.12	2-702549-02	2886	2635	GATE MANUAL-1	ALLOY-718	Alloy 718	19.54	53.12	2.94	5.06	1.00	Yes
2				2636	GATE MANUAL-2	ALLOY-718	Alloy 718	19.40	53.50	3.00	5.12	1.10	Yes
3				2637	GATE MANUAL-3	ALLOY-718	Alloy 718	19.52	53.25	2.98	5.10	0.86	Yes
POST VERIFICATION FOR KNOWN MATERIAL				2688	P.NO : BS625C	Alloy 625	Alloy 625	23.15	60.11	8.58	3.57	0.28	Yes
Notes :													
Tested and Reported By : M.KAJENDRAN NDT LEVEL II				Reviewed By : SURENDRA KUMAR P.V ASNT NDT Level III		Client Reviewed By : INNOVEX		Owner Received By :					
													
Signature Date : 30.03.2026				Signature Date : 31.03.2026		Signature Date :		Signature Date :					

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PMI TESTED AREA



PMI TESTED AREA



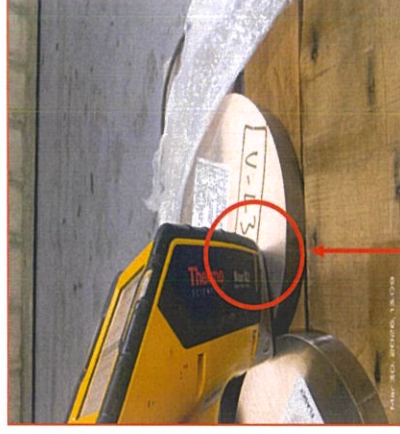
PMI TESTED AREA



PMI TESTED AREA



PMI TESTED AREA



PMI TESTED AREA

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Reading No	Time	Duration	Alloy1	SAMPLE	HEAT	LOT	Mo	Mo Error	Nb	Nb Error	Cu	Cu Error	Ni	Ni Error	Fe	Fe Error	Mn	Mn Error	Cr	Cr Error	Ti	Ti Error
2576	30/03/2026 09:41	11:38	Alloy 625 : 1.53	PRE CAL BS625C	PN-2-702549-02	SO-2886	8.75	0.23	3.58	0.11	<LOD	0.42	60.25	0.61	3.98	0.27	<LOD	0.81	23.26	0.52	0.32	0.85
2635	30/03/2026 12:24	13:03	Alloy 718 : 0.20	GATE MANUAL 2.06inch 1	PN-2-702549-02	SO-2886	2.94	0.07	5.06	0.11	<LOD	0.10	53.12	0.55	18.04	0.35	<LOD	0.40	19.54	0.38	1.00	0.27
2636	30/03/2026 12:25	15:18	Alloy 718 : 0.64	GATE MANUAL 2.06inch 2	PN-2-702549-02	SO-2886	3.00	0.07	5.12	0.10	<LOD	0.41	53.50	0.54	17.82	0.33	<LOD	0.62	19.40	0.36	1.10	0.26
2637	30/03/2026 12:27	13:52	Alloy 718 : 0.68	GATE MANUAL 2.06inch 3	PN-2-702549-02	SO-2886	2.98	0.07	5.10	0.11	<LOD	0.15	53.25	0.55	17.86	0.34	<LOD	0.41	19.52	0.38	0.86	0.26
2688	30/03/2026 14:05	46:30	Alloy 625 : 1.63	POST CAL BS625C			8.58	0.10	3.57	0.05	<LOD	0.23	60.11	0.28	4.13	0.13	<LOD	0.39	23.15	0.25	0.28	0.29

